

Window-frame multiplication

Here's another way of multiplying large numbers together.
Some people find this easier than standard long multiplication.

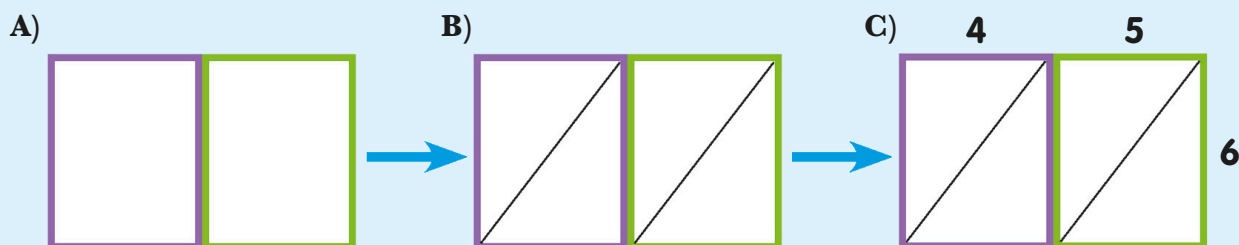
Say, for example, you want to multiply 45 by 6.

A) The number **45** has **2** digits, so draw **2** rectangular boxes side by side.

B) Draw a diagonal line across each box, from the bottom left-hand corner to the top right-hand corner.

C) Write the numbers you want to multiply along the top and right-hand side of the boxes.

The number **6** is only one digit long, so only one row of boxes is needed.



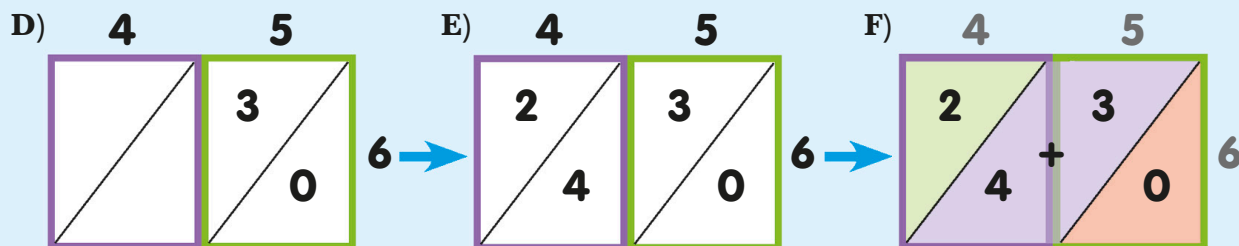
D) Multiply the digits along the top and side, starting from the right.

5 $\times$ **6** = **30**, so write **3** and **0** on either side of the diagonal line.

E) Now do the multiplication in the next box along. **4** $\times$ **6** = **24**, so write **2** and **4**.

F) Look at the numbers in each diagonal column. These give you the answer to **45** $\times$ **6**. If there are two numbers in a diagonal column, add them together.

Add together the numbers in the purple column.



$$45 \times 6 = 270$$